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Inventor(s)	MENG; Fandi

ORAL CARING DEVICE

Abstract

Disclosed is an oral caring device. The oral caring device includes a handle part and a caring head part, the handle part is configured to be suitable for hand-holding, the caring head part includes a flushing stem, and a brush head structure, arranged at the second end of the flushing stem. According to the technical solution of the present disclosure, the oral caring device has a synchronous flushing function to improve a cleaning effect and user experience.

Inventors:	MENG; Fandi (Shenzhen, CN)
Applicant:	Shenzhen Soocas Technology Co., Ltd. (Shenzhen, CN)
Family ID:	83479305
Assignee:	Shenzhen Soocas Technology Co., Ltd. (Shenzhen, CN)
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Background/Summary

CROSSREFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/177,999 titled “ORAL CARING DEVICE”, filed on Mar. 3, 2023, which claims the benefit of Chinese Application No. 202210940250.8, filed on Aug. 5, 2022. The entire disclosure of the above application is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of oral cleaning devices, in particular to an oral caring device.

BACKGROUND

[0003] Oral health is very important for many people. A current oral cleaning mode in the market is generally to rinse an oral cavity with an oral irrigator alone or to brush teeth with an electric toothbrush alone. That is, the current oral irrigator or electric toothbrush is an independent unit, and it is impossible to rinse the oral cavity with a water flow from a brush head at the same time of or after tooth brushing. In this way, in a process of cleaning the oral cavity by a user, the user needs to switch two stems back and forth to provide functions of rinsing and brushing the teeth, which will reduce user experience and reduce a cleaning effect of the oral caring device.

SUMMARY

[0004] The main objective of the present disclosure is to provide an oral caring device. The above oral caring device has a synchronous flushing function to improve a cleaning effect and user experience.

[0005] In order to achieve the above objective, the present disclosure provides an oral caring device, including a handle part and a caring head part. The handle part is configured to be suitable for hand-holding, and the handle part includes: a sonic motor, including a motor body and a power output shaft. The motor body has an output axis extending in a power output direction; the power output shaft is a columnar body extending along the output axis and penetrating through the motor body, the power output shaft is configured to cooperate with the motor body and transmit output power of the motor body, the power output shaft has an axial channel parallel to the output axis, and a first inlet and a first outlet which both communicate with the axial channel, and the first inlet is used for inflow of a fluid on a fluid incoming side. The caring head part includes a flushing stem, defining an flow channel, and a second inlet and a second outlet which both communicate with the flow channel, wherein the flow channel is parallel to the output axis, the second inlet is formed at a first end of the flushing stem, the second outlet is formed at a second end of the flushing stem, the first end of the flushing stem is connected with the power output shaft, and the second inlet communicates with the first outlet; and a brush head structure, arranged at the second end of the flushing stem, wherein the brush head structure includes a brush head body, a plurality of bristles arranged on the brush head body and a water outlet end arranged on the brush head body, the bristles extend in a direction that form a preset included angle with the output axis, and the water outlet end communicates with the second outlet, so that the fluid flows through the axial channel and the flow channel in sequence and then is sprayed out from the water outlet end.

[0006] Further, the brush head body is provided with an installation through hole that forms an included angle with the output axis, the brush head structure further includes a nozzle having an axial through hole, one end of the nozzle communicates with the second outlet, the other end of the nozzle protrudes out of the installation through hole and forms the water outlet end, and the nozzle is installed on the brush head body in a sealed mode; and/or the nozzle is made of an elastic

material.

[0007] Further, the axial through hole includes a first hole segment and a second hole segment which communicate with each other, an inner diameter of the first hole segment is smaller than an inner diameter of the second hole segment, the second hole segment is adjacent to the second outlet and communicates with the second outlet, and the water outlet end is formed at one end of the first hole segment away from the second hole segment.

[0008] Further, along an axis of the axial through hole, at least part of the axial through hole has an inner diameter that gradually decreases along a direction away from the flushing stem, so that the axial through hole has a conical inner surface; and the caring head part further includes a convex part arranged on the flushing stem, the convex part has an inner through hole, the convex part has a conical outer surface, the conical inner surface is in sealed fit with the conical outer surface, and the inner through hole communicates with the second outlet.

[0009] Further, the inner diameter of the first hole segment is greater than or equal to 0.5 mm and less than or equal to 0.7 mm.

[0010] Further, the brush head structure further includes an anti-falling piece arranged on a periphery of the nozzle, the anti-falling piece is located on one side of the brush head body facing away from the bristles, so as to prevent the nozzle from falling out of the installation through hole, and the anti-falling piece is in sealed fit with the brush head body.

[0011] Further, the water outlet end protrudes relative to the brush head body towards one side thereof facing away from the flushing stem, and a distance H1 between an end face of the water outlet end and the brush head body is greater than or equal to 1 mm and less than or equal to 10 mm; and/or, a distance H2 between a top end of the bristles and the brush head body is greater than or equal to 5 mm and less than or equal to 15 mm.

[0012] Further, the second end of the flushing stem is provided with a first installation groove and a second installation groove which communicate with each other, and an interconnection position between the first installation groove and the second installation groove has a first step surface; and the brush head body includes a first segment and a second segment which are interconnected, an interconnection position between the first segment and the second segment has a second step surface, the first segment is located in the first installation groove, at least part of the second segment is located in the second installation groove, and the first step surface is in sealed fit with the second step surface.

[0013] Further, the oral caring device further includes a bracket structure, wherein the bracket structure is configured to be capable of transmitting a driving force of the power output shaft to the flushing stem, the bracket structure is arranged at the first end of the flushing stem, and the bracket structure can be snap-fitted with the power output shaft, so as to limit axial movement of the power output shaft relative to the flushing stem.

[0014] Further, a periphery of the power output shaft is provided with a clamping groove, the clamping groove extends in a circumferential direction of the power output shaft, and the bracket structure includes: a holding bracket, having an inner through hole parallel to the output axis, wherein the power output shaft is inserted into the inner through hole; and a clamping piece, arranged on the holding bracket, wherein the clamping piece includes two clamping arms spaced apart in a radial direction of the power output shaft, and the two clamping arms have a clamping position, in which the two clamping arms are close to each other and are snap-fitted with the clamping groove, and a disengagement position, in which the two clamping arms are away from each other and are disengaged from the clamping groove.

[0015] By applying the technical solution of the present disclosure, compared with using an oral irrigator to clean an oral cavity alone or using an electric toothbrush to clean the oral cavity alone, in the present embodiment, by arranging the axial channel in the power output shaft for power transmission, arranging the flow channel in the flushing stem, and arranging the water outlet end on the brush head body, the fluid may flow through the axial channel and the flow channel in sequence

and then be sprayed out from the water outlet end. In this way, the fluid sprayed out from the water outlet end may further rinse the oral cavity at the same time of or after tooth brushing, so that the oral caring device can not only have a tooth brushing function of the electric toothbrush, but also have a rinsing function of the oral irrigator, thereby improving user experience and improving a cleaning effect of the oral caring device. Therefore, the oral caring device of the present embodiment has a synchronous flushing function.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The specification drawings constituting a part of the present application are configured to provide further understanding to the present disclosure. Schematic embodiments of the present disclosure and their illustration are configured to explain the present disclosure, and do not constitute improper limitation to the present disclosure. In the accompanying drawings:

[0017] FIG. 1 shows a schematic structural diagram of an oral caring device in accordance with an embodiment of the present disclosure;

[0018] FIG. 2 shows a partial enlarged view at A of an oral caring device of FIG. 1;

[0019] FIG. 3 shows a schematic breakdown structural diagram of a caring head part of an oral caring device of FIG. 1;

[0020] FIG. 4 shows a schematic structural diagram of a brush head structure of a caring head part of FIG. 3;

[0021] FIG. 5 shows a partial enlarged view at B of an oral caring device of FIG. 1;

[0022] FIG. 6 shows a schematic breakdown structural diagram of a bracket structure of FIG. 5; and

[0023] FIG. 7 shows a schematic structural diagram of a clamping piece of a bracket structure of FIG. 6.

[0024] Wherein, the above accompanying drawings include the following reference numerals:

[0025] 1. Handle part; 2. Caring head part; 3. Motor body; 4. Power output shaft; 10. Flushing stem; 11. Brush head structure; 12. Bristles; 13. Water outlet end; 14. Axial channel; 15. Flow channel; 16. First inlet; 17. First outlet; 20. Bracket structure; 21. Clamping groove; 22. Holding bracket; 23. Clamping piece; 231. Clamping arm; 232. Elastic arm; 233. Positioning arm; 24. Installation through groove; 25. Stop bar; 26. Positioning bar; 51. Second inlet; 52. Second outlet; 53. Installation through hole; 54. Nozzle; 55. First hole segment; 56. Second hole segment; 57. Convex part; 58. Anti-falling piece; 59. First installation groove; 60. Second installation groove; 61. First segment; and 62. Second segment.

DETAILED DESCRIPTION

[0026] It should be noted that embodiments in the present application and features in the embodiments may be mutually combined in the case of no conflict. The present disclosure will be illustrated below in detail with reference to the accompanying drawings and in combination with the embodiments.

[0027] As shown in FIG. 1, an embodiment of the present disclosure provides an oral caring device. The oral caring device includes a handle part 1 and a caring head part 2. The handle part 1 is configured to be suitable for hand-holding, and the handle part 1 includes a sonic motor. The sonic motor includes a motor body 3 and a power output shaft 4, and the motor body 3 has an output axis extending in a power output direction; the power output shaft 4 is a columnar body extending along the output axis and penetrating through the motor body 3, the power output shaft 4 is configured to cooperate with the motor body 3 and transmit output power of the motor body 3, the power output shaft 4 has an axial channel 14 parallel to the output axis, and a first inlet 16 and a first outlet 17 which both communicate with the axial channel 14, and the first inlet 16 is used for

inflow of a fluid on a fluid incoming side; and the caring head part 2 includes a flushing stem 10 and a brush head structure 11. The flushing stem 10 defines a flow channel 15, and a second inlet 51 and a second outlet 52 which both communicate with the flow channel 15, the flow channel 15 is parallel to the output axis, the second inlet 51 is formed at a first end of the flushing stem 10, the second outlet 52 is formed at a second end of the flushing stem 10, the first end is connected with the power output shaft 4, and the second inlet 51 communicates with the first outlet 17. The brush head structure 11 is arranged at the second end of the flushing stem 10, the brush head structure 11 includes a brush head body, a plurality of bristles 12 arranged on the brush head body and a water outlet end 13 arranged on the brush head body, the bristles 12 extend in a direction that forms a preset included angle with the output axis, and the water outlet end 13 communicates with the second outlet 52, so that the fluid flows through an axial channel 14 and the flow channel 15 in sequence and then is sprayed out from the water outlet end 13.

[0028] In the above technical solution, compared with using an oral irrigator to clean an oral cavity alone or using an electric toothbrush to clean the oral cavity alone, in the present embodiment, by arranging the axial channel 14 in the power output shaft 4 for power transmission, arranging the flow channel 15 in the flushing stem 10, and arranging the water outlet end 13 on the brush head body, the fluid may flow through the axial channel 14 and the flow channel 15 in sequence and then be sprayed out from the water outlet end 13. In this way, the fluid sprayed out from the water outlet end 13 may further rinse the oral cavity at the same time of or after tooth brushing, so that the oral caring device can not only have a tooth brushing function of the electric toothbrush, but also have a rinsing function of the oral irrigator, thereby improving user experience and improving a cleaning effect of the oral caring device. Therefore, the oral caring device of the present embodiment has a synchronous flushing function.

[0029] Further, in the present embodiment, by making the power output shaft 4 have the axial channel 14 parallel to the output axis, and the first inlet 16 and the first outlet 17 communicating with the axial channel 14, when the fluid on the fluid incoming side reaches a position of the sonic motor, it is not necessary to run a pipe outside the sonic motor, and it can be directly connected with the power output shaft 4 of the sonic motor, so that the axial channel 14 of the power output shaft 4 can be utilized for fluid transportation, and thus the fluid can flow through the interior of the power output shaft 4 inside the sonic motor without making other changes in a product structure (that is, avoiding additionally arranging a transmission pipeline to transport the fluid) and without occupying an additional space, and the structure of the sonic motor can be utilized more fully and reasonably, thereby making structural layout of a rinsing flow path more reasonable, reducing a fluid pipeline, saving space occupation, and making an overall size of the oral caring device smaller, a weight lighter, and the stem part more convenient to hold.

[0030] It should be noted that the axial channel 14 of the present application can not only convey a water flow, but also convey toothpaste or rinsing fluid.

[0031] As shown in FIG. 2 and FIG. 3, in the embodiment of the present disclosure, the brush head body is provided with an installation through hole 53 that forms an included angle with the output axis, the brush head further includes a nozzle 54 having an axial through hole, one end of the nozzle 54 communicates with the second outlet 52, the other end of the nozzle 54 protrudes out of the installation through hole 53 and forms the water outlet end 13, and the nozzle 54 is installed on the brush head body in a sealed mode.

[0032] In the above technical solution, compared with directly making the fluid flow out of the installation through hole 53, in the present embodiment, the water flowing out of the second outlet 52 may be sprayed out from the nozzle 54 by arranging the nozzle 54 penetrating out of the installation through hole 53. In this way, a distance between the water outlet end 13 and teeth can be reduced, so that the water outlet end 13 is closer to the teeth to flush the teeth better, so as to improve the cleaning effect of the oral caring device. It should be noted that the central axis of the nozzle 54 may be perpendicular to the brush head body, or may have an included angle of an acute

angle of less than 90° with the brush head body.

[0033] Preferably, in the embodiment of the present disclosure, the nozzle **54** is made of an elastic material, preferably rubber. In this way, on the one hand, when brushing the teeth, if the nozzle **54** makes contact with the teeth due to a strong force of a user, a problem of poor experience caused by the nozzle **54** hitting the teeth can be avoided. On the other hand, the nozzle **54** is used to be installed on a toothbrush, which has a delicate structure and is made of the elastic material for easy machining, so as to avoid a problem of cracks when machining the axial through hole. In addition, some technologies use additional copper installation pieces to assemble the nozzle **54**. After a period of use, the copper installation pieces are prone to producing copper rust, which will affect oral health, and cause poor user experience. In the embodiment of the present application, the nozzle **54** made of the rubber is directly installed in the installation through hole **53** without additionally arranging the copper installation pieces, thereby improving the use experience of the user.

[0034] As known to the inventor, the brush head structure generally adopts copper bristle-planting, which is prone to producing the copper rust after tooth brushing, especially after rinsing, and affects the oral health. Therefore, a bristle planting process of the brush head structure **11** in the present embodiment adopts a copper-free bristle planting process.

[0035] As shown in FIG. 2, in the embodiment of the present application, the axial through hole includes a first hole segment **55** and a second hole segment **56** which communicate with each other, an inner diameter of the first hole segment **55** is smaller than an inner diameter of the second hole segment **56**, the second hole segment **56** is adjacent to the second outlet **52** and communicates with the second outlet **52**, and the water outlet end **13** is formed at one end of the first hole segment **55** away from the second hole segment **56**.

[0036] Through the above arrangement, after flowing out of the second outlet **52**, the fluid may be sprayed out from the second hole segment **56** and the first hole segment **55** in sequence. Since the inner diameter of the first hole segment **55** is smaller than the inner diameter of the second hole segment **56**, when the flow quantity is constant, a flow rate of the fluid in the first hole segment **55** is greater than a flow rate in the second hole segment **56**. In this way, a spraying-out speed of the fluid may be increased, thus increasing the cleaning effect of the fluid on the teeth.

[0037] As shown in FIG. 2, in the embodiment of the present disclosure, along an axis of the axial through hole, at least part of the axial through hole has an inner diameter that gradually decreases along a direction away from the flushing stem **10**, so that the axial through hole has a conical inner surface; and the caring head part **2** further includes a convex part **57** arranged on the flushing stem **10**, the convex part **57** has an inner through hole, the convex part **57** has a conical outer surface, the conical inner surface is in sealed fit with the conical outer surface, and the inner through hole communicates with the second outlet **52**.

[0038] In the above technical solution, on the one hand, radial seal may be realized by arranging the convex part **57** which is in sealed fit with the axial through hole, so as to prevent the fluid in the second hole segment **56** from flowing out of a position between the nozzle **54** and the brush head body; on the other hand, an effect of axial water blocking may be increased by arranging the conical inner surface and the conical outer surface which are in sealed fit, so as to prevent the fluid in the second hole segment **56** from flowing out of a position between an outer wall surface of the convex part **57** and an inner wall surface of the second hole segment **56**, thereby increasing a sealing effect to improve the user experience.

[0039] Further, the nozzle **54** made of the elastic material may be expanded on the periphery of the convex part **57** under the elastic force to increase the sealing effect.

[0040] Preferably, in the embodiment of the present disclosure, the inner diameter of the second hole segment **56** gradually decreases along the direction away from the flushing stem **10**, and the second hole segment **56** has the conical inner surface cooperating with the conical outer surface. Of course, in an alternative embodiment not shown in the accompanying drawings, an inner diameter

of the entire axial through hole may also gradually decrease as long as the conical inner surface can be formed.

[0041] Preferably, in the embodiment of the present disclosure, the inner diameter of the first hole segment is greater than or equal to 0.5 mm and less than or equal to 0.7 mm. In this way, the speed at which the fluid is sprayed out from the first hole segment 55 may be effectively increased, thereby increasing cleaning force of the fluid on the oral cavity.

[0042] As shown in FIG. 2 and FIG. 3, in the embodiment of the present disclosure, the brush head structure further includes an anti-falling piece 58 arranged on a periphery of the nozzle 54, the anti-falling piece 58 is located on one side of the brush head body facing away from the bristles 12, so as to prevent the nozzle 54 from falling out of the installation through hole 53, and the anti-falling piece 58 is in sealed fit with the brush head body.

[0043] In the above technical solution, when flowing in the axial through hole of the nozzle 54, the fluid will produce an axial impact force on the nozzle 54. By arranging the anti-falling piece 58 located on one side of the brush head body facing away from the bristles 12, the anti-falling piece 58 may be limited between the brush head body and the flushing stem 10, which can prevent the nozzle 54 from falling out of the installation through hole 53, so as to more stably fix the nozzle 54. Thus, the oral caring device has the functions of rinsing the oral cavity and brushing the teeth at the same time to improve the user experience.

[0044] Preferably, the anti-falling piece 58 and the nozzle 54 are of an integrally-formed structure, and the anti-falling piece is also made of the elastic materials such as rubber.

[0045] Specifically, in the embodiment of the present disclosure, the brush head body is pressed-fit on the anti-falling piece 58 to achieve the sealed fit between the anti-falling piece 58 and the brush head body. In this way, the sealing effect may be increased to prevent the fluid in the nozzle 54 from flowing out of the position between the nozzle 54 and the brush head body.

[0046] As shown in FIG. 2, in the embodiment of the present disclosure, the water outlet end 13 protrudes relative to the brush head body towards one side thereof facing away from the flushing stem 10, and a distance H1 between an end face of the water outlet end 13 and the brush head body is greater than or equal to 1 mm and less than or equal to 10 mm, preferably, H is greater than or equal to 2 mm and less than or equal to 8 mm; and/or, a distance H2 between a top end of the bristles 12 and the brush head body is greater than or equal to 5 mm and less than or equal to 15 mm.

[0047] Through the above arrangement, not only can the water outlet end 13 be close enough to the teeth so that the sprayed fluid has the good cleaning effect, but also the probability of the water outlet end 13 in contact with the teeth can be reduced in the process of tooth brushing, so that discomfort caused by the water outlet end 13 in contact with and rubbing with the teeth can be reduced, thereby improving the user experience.

[0048] Specifically, in the embodiment of the present disclosure, a ratio of the distance H1 to the distance H2 is greater than or equal to 0.25 and less than or equal to 0.5. In this way, while ensuring that the fluid has sufficient cleaning power, the probability of contact between the water outlet end 13 and the teeth can further be reduced, thereby improving the user experience.

[0049] As shown in FIG. 3 and FIG. 4, in the embodiment of the present disclosure, the second end of the flushing stem 10 is provided with a first installation groove 59 and a second installation groove 60 which communicate with each other, and an interconnection position between the first installation groove 59 and the second installation groove 60 has a first step surface; and the brush head body includes a first segment 61 and a second segment 62 which are interconnected, an interconnection position between the first segment 61 and the second segment 62 has a second step surface, the first segment 61 is located in the first installation groove 59, at least part of the second segment 62 is located in the second installation groove 60, and the first step surface is in sealed fit with the second step surface.

[0050] Through the above arrangement, not only can the nozzle 54 be pressed-fit on the flushing

stem **10** through the brush head body, but also the sealing between the anti-falling piece **58** and the brush head body, the sealing between the anti-falling piece **58** and the flushing stem **10**, and the sealing between the brush head body and the flushing stem **10** can be realized, so as to prevent water from flowing out from the axial through hole of the nozzle **54** via a position between the anti-falling piece **58** and the brush head body, or between the anti-falling piece **58** and the flushing stem **10**, or between the brush head body and the flushing stem **10**, thereby further avoiding reducing the user experience.

[0051] Specifically, in the embodiment of the present disclosure, the brush head body is pressed-fit on the flushing stem **10** in an ultrasonic pressing-fit mode, and the first step surface and the second step surface are ultrasonically bound to increase the sealing effect.

[0052] As shown in FIG. **1**, in the embodiment of the present disclosure, the oral caring device further includes a bracket structure **20**. The bracket structure **20** is configured to be capable of transmitting a driving force of the power output shaft **4** to the flushing stem **10**, the bracket structure **20** is arranged at the first end of the flushing stem **10**, and the bracket structure **20** can be snap-fitted with the power output shaft **4**, so as to limit axial displacement of the power output shaft **4** relative to the flushing stem **10**.

[0053] In the above technical solution, compared with a problem that the electric toothbrush in the prior art only needs to meet the requirements of vibration transmission or the oral irrigator only needs to consider hydraulic impact, the present embodiment can not only make the bracket structure **20** transmit the driving force of the power output shaft **4** to the flushing stem **10** by arranging the bracket structure **20**, so as to ensure that the vibration is transmitted to the flushing stem **10**, thus realizing the tooth brushing function of the electric toothbrush. Moreover, the bracket structure **20** is snap-fitted with the power output shaft **4**, which can further limit the axial displacement of the power output shaft **4** relative to the flushing stem **10**. In this way, a problem that the power output shaft **4** falls out of the flushing stem **10** due to the hydraulic impact can be avoided, so that swinging force transmitted by the power output shaft **4** can be more stably transmitted to the flushing stem **10**, thus making the oral caring device in the present embodiment not only have the tooth brushing function of the electric toothbrush, but also have the flushing function of the oral irrigator, thereby improving the user experience and improving the cleaning effect of the oral caring device. Therefore, the oral caring device of the present embodiment has a synchronous flushing function.

[0054] It should be noted that in the embodiment of the present disclosure, when the fluid flows from the power output shaft **4** into the flow channel **15** of the flushing stem **10**, the high-pressure fluid will produce a water flow impact force on the end of the power output shaft **4**, which is prone to causing the power output shaft **4** in FIG. **1** to fall out of the flushing stem **10**. However, the present embodiment can solve this problem by arranging the bracket structure **20**.

[0055] As shown in FIG. **1**, FIG. **5**, FIG. **6** and FIG. **7**, in the embodiment of the present disclosure, a periphery of the power output shaft **4** is provided with a clamping groove **21**, the clamping groove **21** extends in a circumferential direction of the power output shaft **4**, and the bracket structure **20** includes a holding bracket **22** and a clamping piece **23**. The holding bracket **22** have an inner through hole parallel to the output axis, and the power output shaft **4** is inserted into the inner through hole; and the clamping piece **23** is arranged on the holding bracket **22**, the clamping piece **23** includes two clamping arms **231** spaced apart in a radial direction of the power output shaft **4**, and the two clamping arms **231** have a clamping position, in which the two clamping arms **231** are close to each other and are snap-fitted with the clamping groove, and a disengagement position, in which the two clamping arms **231** are away from each other and are disengaged from the clamping groove **21**.

[0056] Through the above arrangement, when the power output shaft **4** is inserted into the inner through hole and a position between the two clamping arms **231**, the two clamping arms **231** may be away from each other under the action of the power output shaft **4** until the clamping groove **21**

of the power output shaft **4** moves to a position corresponding to the clamping arms **231**. The two clamping arms **231** will be close to each other and are clamped into the clamping groove **21**, so as to limit the axial displacement of the power output shaft **4** relative to the holding bracket **22**. In this way, the problem that the power output shaft **4** falls out of the flushing stem **10** due to the hydraulic impact may be avoided, so that the swinging force transmitted by the power output shaft **4** can be more stably transmitted to the flushing stem **10**, and thus the oral caring device in the present embodiment not only has the tooth brushing function of the electric toothbrush, but also has the flushing function of flushing the teeth.

[0057] As shown in FIG. 7, in the embodiment of the present disclosure, the clamping piece **23** further includes an elastic arm **232** for connecting the two clamping arms **231**, so that the two clamping arms **231** are switched from the disengagement position to the clamping position.

[0058] Through the above arrangement, when the power output shaft **4** is installed into the inner through hole, under the action of assembly force of the power output shaft **4**, the two clamping arms **231** may be opened, the elastic arm **232** deforms until the clamping groove **21** of the power output shaft **4** moves to the position corresponding to the clamping arms **231**, and then the elastic arm **232** recovers its deformation to drive the two clamping arms **231** to get close to each other, so as to be clamped into the clamping groove **21**, which prevents the axial movement of the power output shaft **4** relative to the holding bracket **22**. In this way, the problem that the power output shaft **4** falls out of the flushing stem **10** due to the hydraulic impact may be avoided.

[0059] Further, since the elastic arm **232** is elastic, when it is necessary to remove the power output shaft **4** from the flushing stem **10**, a user may use a force greater than the impact force of a water flow to pull the power output shaft **4** out of the inner through hole. In this way, the elastic arm **232** will deform to drive the two clamping arms **231** away from each other, so that the clamping arms **231** are disengaged from the clamping groove **21**. In this way, when the brush head structure **11** needs to be replaced due to wear after a long time of use, the above structure may realize rapid disassembly, which is convenient for the user to timely replace the caring head part **2**.

[0060] Preferably, in the embodiment of the present disclosure, an inner wall of the clamping groove **21** is connected with a side wall of the power output shaft **4** through an arc-shaped transition section. In this way, when the power output shaft **4** is installed, the clamping arms **231** may slide more smoothly into the clamping groove **21** from the side wall of the power output shaft **4**, and when the power output shaft **4** is disassembled, the clamping arms **231** may slide more smoothly out of the interior of the clamping groove **21**.

[0061] In an alternative embodiment, only two clamping arms **231** may be arranged, and are made of the elastic material. In this way, under the action of the power output shaft **4**, the two clamping arms **231** may be away from each other, and when the clamping groove **21** of the power output shaft **4** moves to the position corresponding to the clamping arms **231**, the two clamping arms **231** will recover deformation and get close to each other to be clamped into the clamping groove **21**.

[0062] As shown in FIG. 1, FIG. 5 and FIG. 6, in the embodiment of the present disclosure, the holding bracket **22** is provided with an installation through groove **24** communicating with the inner through hole. The installation through groove **24** runs through the holding bracket **22** in the radial direction of the power output shaft **4**; and a circumferential inner wall of the installation through groove **24** is in clearance fit with the clamping arms **231**.

[0063] Through the above arrangement, the clamping piece **23** may be installed in the installation through groove **24**, and the two clamping arms **231** may be exposed from the inner through hole and clamped with the clamping groove **21**. Further, the circumferential inner wall of the installation through groove **24** is in clearance fit with the clamping arms **231**, so that the circumferential inner wall of the installation through groove **24** may limit ultimate displacement of the clamping arms **231**, so as to avoid the failure of the elastic arm **232** due to large deformation when the two clamping arms **231** are away from each other.

[0064] Specifically, in the embodiment of the present disclosure, a width of the installation through

groove **24** on a horizontal plane in FIG. **6** is greater than the inner diameter of the inner through hole, so that the circumferential inner wall of the installation through groove **24** may be in clearance fit with the clamping arms **231**.

[0065] As shown in FIG. **5** and FIG. **7**, in the embodiment of the present disclosure, the clamping piece **23** further includes the elastic arm **232** for connecting the two clamping arms **231**, the bracket structure **20** further includes a stop bar **25** connected with the holding bracket **22**, one end of the installation through groove **24** is provided with the stop bar **25**, and part of the elastic arm **232** is located on one side of the stop bar **25** facing away from the inner through hole (that is, located on the outer side of the stop bar **25**) to limit the displacement of the elastic arm **232**.

[0066] In the above technical solution, when the clamping piece **23** is installed in the installation through groove **24**, the stop bar **25** may limit the displacement of the elastic arm **232**, so as to prevent the elastic arm **232** from being brought into the inner through hole because the power output shaft **4** is inserted into or out of the inner through hole, so as to prevent the elastic arm **232** entering the inner through hole from affecting the disassembly and installation of the power output shaft **4**.

[0067] As shown in FIG. **5** and FIG. **7**, in the embodiment of the present disclosure, the bracket structure **20** further includes a positioning bar **26** arranged at the other end of the installation through groove **24**, the clamping piece **23** further includes two positioning arms **233** connected with the two clamping arms **231** respectively, and the two positioning arms **233** are spaced and located on two sides of the positioning bar **26**.

[0068] Through the above arrangement, on the one hand, the positioning bar **26** may position the two positioning arms **233**. In the disassembly and installation process of the power output shaft **4**, the positioning bar **26** may make the clamping piece **23** located at the same position of the installation through groove **24**, thus preventing the position of the clamping piece **23** from shifting due to the power output shaft **4** penetrating into or out of the inner through hole, so that the power output shaft **4** can be disassembled and installed conveniently and quickly. On the other hand, the positioning bar **26** can open the two positioning arms **233**, so that a spacing between the two clamping arms **231** can make the power output shaft **4** inserted therein.

[0069] Preferably, in the embodiment of the present embodiment, the stop bar **25** and the positioning bar **26** are symmetrically arranged relative to an axis of the inner through hole of the holding bracket **22**. In this way, any one of the positioning bar **26** and the stop bar **25** may limit the displacement of the elastic arm **232** or position the clamping piece **23**, which is convenient for machining.

[0070] Preferably, in the embodiment of the present disclosure, the clamping piece **23** is a U-shaped clamp spring, which certainly can also be a spring of other forms.

[0071] In one embodiment, the oral caring device further includes a fluid mechanism and a battery. The fluid mechanism includes a flow-in inlet that can transport the incoming fluid to the fluid mechanism and a flow-out outlet that can transport the incoming fluid to the first inlet **16**. The fluid mechanism includes a power plant that can provide driving force for flowing of the fluid, so that the fluid flows from the flow-in inlet to the flow-out outlet, and the fluid may pass through the fluid mechanism, the axial channel **14** and the flow channel **15** in sequence and then is sprayed out from the water outlet end **13**. Preferably, the power plant may be a gear pump or a diaphragm pump, etc.

[0072] The battery can provide an energy source for the power plant and the sonic motor. The battery is arranged at one end of the fluid mechanism away from the sonic motor. The sonic motor, the fluid mechanism and the battery are arranged in sequence in a direction parallel to the output axis.

[0073] In the present embodiment, under the driving of the power plant of the fluid mechanism, after flowing out of a water storage cavity, the fluid enters the fluid mechanism through the flow-in inlet of the fluid mechanism, flows out of the flow-out outlet of the fluid mechanism, then is transported to the power output shaft **4**, enters the axial channel **14** from the first inlet **16** of the

axial channel **14** of the power output shaft **4**, then flows from the axial channel **14** to the other end of the motor body **3**, flows out from the first outlet **17** of the axial channel **14**, enters the caring head part **2**, and is finally sprayed out from the water outlet end **13** of the caring head part **2** to rinse the oral cavity. In addition, the sonic motor can output power to the caring head part **2** through the power output shaft **4**, and transmit it to the brush head structure **11** through the flushing stem **10** of the caring head part **2**, so that the brush head structure **11** can brush under the action of the sonic motor, and the oral cavity is cleaned by utilizing the bristles **12** of the brush head structure **11**.

[0074] It can be seen from the above description that the above embodiments of the present disclosure achieve the following technical effects: compared with using the oral irrigator to clean the oral cavity alone or using the electric toothbrush to clean the oral cavity alone, in the present embodiment, by arranging the axial channel in the power output shaft for power transmission, arranging the flow channel in the flushing stem, and arranging the water outlet end on the brush head body, the fluid may flow through the axial channel and the flow channel in sequence and then be sprayed out from the water outlet end. In this way, the fluid sprayed out from the water outlet end may further rinse the oral cavity at the same time of or after tooth brushing, so that the oral caring device can not only have the tooth brushing function of the electric toothbrush, but also have the rinsing function of the oral irrigator, thereby improving the user experience and improving the cleaning effect of the oral caring device. Therefore, the oral caring device of the present embodiment has the synchronous flushing function.

[0075] The above descriptions are only preferred embodiments of the present disclosure, and are not intended to limit the present disclosure. For those skilled in the art, the present disclosure can have various changes and variations. Any modification, equivalent replacement, improvement and the like made within the spirit and scope of the present disclosure shall all be included in the protection scope of the present disclosure.

Claims

1. An oral caring device, comprising a handle part and a caring head part, wherein the handle part is configured to be suitable for hand-holding, and the handle part comprises: a sonic motor, comprising a motor body and a power output shaft, wherein the motor body has an output axis extending in a power output direction; the power output shaft is a columnar body extending along the output axis and penetrating through the motor body, the power output shaft is configured to cooperate with the motor body and transmit output power of the motor body, the power output shaft has an axial channel parallel to the output axis, and a first inlet and a first outlet which both communicate with the axial channel, and the first inlet is used for inflow of a fluid on a fluid incoming side; and the caring head part comprises: a flushing stem, defining a flow channel, and a second inlet and a second outlet which both communicate with the flow channel, wherein the flow channel is parallel to the output axis, the second inlet is formed at a first end of the flushing stem, the second outlet is formed at a second end of the flushing stem, the first end of the flushing stem is connected with the power output shaft, and the second inlet communicates with the first outlet; and a brush head structure, arranged at the second end of the flushing stem, wherein the brush head structure comprises a brush head body, a plurality of bristles arranged on the brush head body and a water outlet end arranged on the brush head body, the bristles extend in a direction that forms a preset included angle with the output axis, and the water outlet end communicates with the second outlet, so that the fluid flows through the axial channel and the flow channel in sequence and then is sprayed out from the water outlet end.

2. The oral caring device according to claim 1, wherein the brush head body is provided with an installation through hole that forms an included angle with the output axis, the brush head structure further comprises a nozzle having an axial through hole, one end of the nozzle communicates with the second outlet, another end of the nozzle protrudes out of the installation through hole and forms

the water outlet end, and the nozzle is installed on the brush head body in a sealed mode; and/or the nozzle is made of an elastic material.

3. The oral caring device according to claim 2, wherein the axial through hole comprises a first hole segment and a second hole segment which communicate with each other, an inner diameter of the first hole segment is smaller than an inner diameter of the second hole segment, the second hole segment is adjacent to the second outlet and communicates with the second outlet, and the water outlet end is formed at one end of the first hole segment away from the second hole segment.

4. The oral caring device according to claim 2, wherein, along an axis of the axial through hole, at least part of the axial through hole has an inner diameter that gradually decreases along a direction away from the flushing stem, so that the axial through hole has a conical inner surface; and the caring head part further comprises a convex part arranged on the flushing stem, the convex part has an inner through hole, the convex part has a conical outer surface, the conical inner surface is in sealed fit with the conical outer surface, and the inner through hole communicates with the second outlet.

5. The oral caring device according to claim 3, wherein the inner diameter of the first hole segment is greater than or equal to 0.5 mm and less than or equal to 0.7 mm.

6. The oral caring device according to claim 2, wherein the brush head structure further comprises an anti-falling piece arranged on a periphery of the nozzle, the anti-falling piece is located on one side of the brush head body facing away from the bristles, so as to prevent the nozzle from falling out of the installation through hole, and the anti-falling piece is in sealed fit with the brush head body.

7. The oral caring device according to claim 1, wherein the water outlet end protrudes relative to the brush head body towards one side thereof facing away from the flushing stem, and a distance H1 between an end face of the water outlet end and the brush head body is greater than or equal to 1 mm and less than or equal to 10 mm; and/or, a distance H2 between a top end of the bristles and the brush head body is greater than or equal to 5 mm and less than or equal to 15 mm.

8. The oral caring device according to claim 1, wherein the second end of the flushing stem is provided with a first installation groove and a second installation groove which communicate with each other, and an interconnection position between the first installation groove and the second installation groove has a first step surface; and the brush head body comprises a first segment and a second segment which are interconnected, an interconnection position between the first segment and the second segment has a second step surface, the first segment is located in the first installation groove, at least part of the second segment is located in the second installation groove, and the first step surface is in sealed fit with the second step surface.

9. The oral caring device according to claim 1, further comprising a bracket structure, wherein the bracket structure is configured to be capable of transmitting a driving force of the power output shaft to the flushing stem, the bracket structure is arranged at the first end of the flushing stem, and the bracket structure can be snap-fitted with the power output shaft, so as to limit axial movement of the power output shaft relative to the flushing stem.

10. The oral caring device according to claim 9, wherein a periphery of the power output shaft is provided with a clamping groove, the clamping groove extends in a circumferential direction of the power output shaft, and the bracket structure comprises: a holding bracket, having an inner through hole parallel to the output axis, wherein the power output shaft is inserted into the inner through hole; and a clamping piece, arranged on the holding bracket, wherein the clamping piece comprises two clamping arms spaced apart in a radial direction of the power output shaft, and the two clamping arms have a clamping position, in which the two clamping arms are close to each other and are snap-fitted with the clamping groove, and a disengagement position, in which the two clamping arms are away from each other and are disengaged from the clamping groove.
